

Depression Risk Profiling with KNHANES & Air-Pollution Data - Public Excerpt

Public-safe technical excerpt derived from the original project output.

Source: 2025 healthcare big-data analysis competition report

Analytical dataset

14,773 observations and 71 variables after data integration, validity checks, missing-value handling, encoding, and scaling.

Pipeline

KNHANES + air-pollution integration; K-Means subgrouping (K=5); SMOTE; cluster-specific Random Forest modeling; feature-importance interpretation.

Key insight

Depression-risk factors differed across clusters, supporting a heterogeneous subgroup-oriented interpretation rather than a single population-wide risk profile.

Public release scope

Underlying KNHANES and environmental datasets are not redistributed; users should obtain them from the original providers.